

From Inspection to Intervention: Causal Mechanisms and Algorithmic Understanding in Neural Networks

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Abstract

Recent advances in mechanistic interpretability have shifted attention from descriptive inspection of neural networks toward causal analysis of their internal operations. This essay synthesizes a causal paradigm in which mechanistic understanding is defined not by access to internal states, but by the ability to perform interventions, evaluate counterfactuals, and recover algorithmic structure through interventional equivalence. We examine the progression from activation-level analysis to causal mediation and abstraction, and we argue that genuine understanding requires sensitivity to history, irreversibility, and responsibility rather than mere correlational control. The resulting framework reframes interpretability as a problem of reverse-engineering algorithms implemented by dynamical systems under constraint.

1 Introduction

The problem of understanding neural networks has long been framed as one of visibility. If the internal weights, activations, or attention patterns could be made sufficiently transparent, then the systems behavior would become intelligible. This intuition has motivated a wide range of visualization and probing techniques that aim to “open the black box.”

However, visibility alone does not constitute understanding. Observing that a component activates in the presence of a behavior does not establish that the component is responsible for that behavior, nor does it explain how the behavior would change under alternative conditions. The gap between observation and explanation motivates a more demanding notion of mechanistic understanding, one rooted in causality rather than correlation.

This essay develops such a notion. Mechanistic understanding is taken to mean the ability to identify internal components whose manipulation predictably and systematically alters behavior in a way that corresponds to an intelligible algorithm. On this view, interpretability becomes a causal science.

2 What It Means to Be Mechanistic

The term “mechanistic” is often used ambiguously. In its strongest technical sense, a mechanistic explanation specifies not merely which parts are present in a system, but how those parts causally interact to produce behavior.

Definition 1 (Mechanistic Understanding). *A system is mechanistically understood with respect to a behavior if one can identify a set of internal variables such that interventions on those variables produce predictable, structured changes in the behavior, consistent with an underlying algorithmic description.*

This definition excludes purely observational accounts. A neuron that reliably activates during correct answers may be diagnostically useful, but it is not mechanistically explanatory unless altering its value changes the outcome in a lawful way.

3 Causality and Intervention

Causality enters the picture through intervention. An intervention sets an internal variable to a value it would not ordinarily take, breaking its normal dependencies and revealing its downstream effects.

Formally, an intervention replaces the natural evolution of a variable X with an assignment $\text{do}(X = x)$, holding fixed the rest of the systems causal structure. The resulting behavior distinguishes genuine causal influence from mere correlation.

Counterfactual reasoning extends this idea. Rather than asking what happens when a variable is zeroed or scaled, one asks what would have happened had the system internally evolved as if it were processing a different input. Such counterfactuals probe whether a component carries information that is functionally responsible for a computation.

4 Activation-Level Control

One of the earliest demonstrations of causal influence comes from activation steering. By adding a vector to internal activations, researchers can bias a model toward or away from certain behaviors. When steering vectors are derived from differences in mean activation between contrasting behaviors, they reveal that behavioral distinctions often correspond to approximately linear directions in representation space.

Despite its usefulness, activation steering remains limited. It demonstrates that internal representations are manipulable, but it does not by itself specify what computation is being performed. Control without explanation is not yet mechanistic understanding.

5 Causal Mediation and Information Flow

Causal mediation analysis refines intervention by decomposing the total effect of an input on an output into contributions carried by intermediate variables. Rather than asking whether a

component matters, mediation asks how and where information flows.

In causal tracing experiments, noise is injected to degrade a models performance, after which selected components are restored to their clean values. Components whose restoration recovers correct behavior are identified as mediators. This procedure reveals structured pathways by which information is retrieved, transformed, and routed through the network.

Such analyses produce causal stories rather than static maps. They show not only where information is represented, but how it moves.

6 Causal Abstraction and Algorithmic Structure

The strongest form of mechanistic interpretability arises when neural activity can be related to a higher-level algorithm through causal abstraction. Both the neural network and the algorithm are treated as causal models, and the goal is to determine whether the algorithm faithfully abstracts the networks behavior under intervention.

Definition 2 (Causal Abstraction). *A high-level model $\mathcal{A}$ is a causal abstraction of a low-level system if there exists a mapping between their variables such that corresponding interventions produce equivalent effects on outputs.*

Interchange interventions play a central role here. Instead of modifying a variable arbitrarily, one replaces its value with the value it would have taken under a different input. If such substitutions produce output changes that match the predictions of the abstract algorithm, the abstraction is validated.

This criterion is stringent. It requires not just similarity of behavior, but interventional equivalence.

7 Events, History, and Irreversibility

Causal abstraction implicitly raises questions about the role of history. Two internal states may be numerically identical yet differ in how they were produced. If their downstream effects diverge under intervention, then the mechanism is history-sensitive.

This suggests that mechanistic explanation may require attention not only to states, but to events: transformations that irreversibly constrain future behavior. Interventions that cannot be undone without loss of explanatory power signal the presence of genuine historical commitment within the system.

8 Responsibility and Constraint

Not all mechanisms act by producing outcomes. Some function by suppressing, filtering, or excluding possibilities. Such negative or constraining roles are often invisible to activation-based analyses, yet they may be causally decisive.

A component may bear responsibility for a behavior not by optimizing for it, but by preventing alternatives. Recognizing such mechanisms requires a notion of causality that treats absence and inhibition as first-class explanatory factors.

9 Limits of Understanding

Successful interventions do not automatically constitute understanding. A collection of techniques may enable control without yielding a coherent explanatory model. Conversely, an elegant abstraction may fail outside a narrow regime of validity.

Mechanistic understanding therefore admits of degrees. It improves as abstractions become more stable under distribution shift, more sensitive to history, and more faithful under counterfactual substitution.

10 Conclusion

Mechanistic interpretability, when grounded in causality, becomes the study of which internal structures are responsible for behavior and which algorithms they realize. This shift moves the field beyond visualization and toward reverse-engineering.

The central lesson is that understanding is not achieved by looking, but by intervening; not by correlating, but by substituting; and not by controlling outputs alone, but by reconstructing the causal logic that produces them. Only then can claims about what a neural network is doing rise from metaphor to mechanism.